LEGO Design Enrichment Program - Term 2

8-Week Curriculum Guide

Think. Play. Grow.

Beginner Level

Expanding creativity: Thematic building, color theory, and collaborative projects

Week 1: Underwater World

Component	Description
Learning Goals	• Build sea creatures and submarines • Use blue and green effectively • Create underwater scenes
Activities	• Fish and shark building • Submarine design challenge • Coral reef collaboration
Superpower	Theme Development - Building cohesive worlds

Week 2: Space Exploration

Component	Description
Learning Goals	• Design rockets and spaceships • Build alien creatures • Create space stations
Activities	• Rocket launch competition • Alien design contest • Space station module building
Superpower	Futuristic Thinking - Imagining tomorrow

Week 3: Medieval Castles

Component	Description
Learning Goals	• Build castle structures • Create knights and dragons • Add defensive features

Component	Description
Activities	• Castle tower building • Drawbridge mechanics • Knight and dragon battles
Superpower	Historical Imagination - Building the past

Week 4: Robots & Automatons

Component	Description
Learning Goals	• Design robot characters • Add movable joints • Create robot functions
Activities	• Robot design challenge • Articulated joint practice • Robot story creation
Superpower	Mechanical Thinking - Making things move

Week 5: Nature & Gardens

Component	Description
Learning Goals	• Build trees and flowers • Create park scenes • Use natural colors effectively
Activities	• Botanical garden creation • Tree building techniques • Landscape design
Superpower	Organic Building - Curved and natural forms

Week 6: Sports Stadium

Component	Description
Learning Goals	• Build sports venues • Create playing fields • Add spectator areas
Activities	• Stadium construction • Field design (soccer, baseball) • Scoreboard building

Component	Description
Superpower	Scale Building - Large structures

Week 7: Color Mastery

Component	Description
Learning Goals	• Understand color combinations • Build monochromatic creations • Use color for emphasis
Activities	• Rainbow building challenge • Single-color masterpieces • Color accent exercises
Superpower	Artistic Eye - Using color effectively

Week 8: Gallery Exhibition

Component	Description
Learning Goals	• Curate personal collection • Present creations • Appreciate others' work
Activities	• Gallery setup • Presentation practice • Exhibition celebration
Superpower	Presentation - Showcasing work

Intermediate Level

Technical advancement: Gear trains, advanced structures, and design challenges

Week 1: Gear Systems

Component	Description
Learning Goals	• Build multi-stage gear trains • Understand gear ratios • Create gear-driven machines
Activities	• Gear ratio experiments • Mechanical advantage challenges • Gear-powered vehicles
Superpower	Mechanical Engineering - Power transmission

Week 2: Bridge Engineering

Component	Description
Learning Goals	• Design different bridge types • Test load capacity • Optimize structure-to-weight ratio
Activities	• Arch bridge construction • Suspension bridge challenges • Weight-bearing tests
Superpower	Engineering Analysis - Testing and improving

Week 3: Technic Building

Component	Description
Learning Goals	• Use Technic pieces effectively • Build sturdy frameworks • Create functional models
Activities	• Technic frame construction • Lifting mechanisms • Heavy-duty vehicle building
Superpower	Technical Building - Using specialized pieces

Week 4: Rube Goldberg Machines

Component	Description
Learning Goals	• Design chain reaction machines • Connect multiple actions • Create entertaining contraptions
Activities	• Simple chain reactions • Multi-step machines • Rube Goldberg competition
Superpower	Sequential Thinking - One thing leads to another

Week 5: Architecture Styles

Component	Description
Learning Goals	• Study architectural periods • Build in different styles • Understand design elements
Activities	• Gothic architecture • Modern minimalist design • Frank Lloyd Wright inspired
Superpower	Architectural Knowledge - Understanding styles

Week 6: Lighting & Effects

Component	Description
Learning Goals	• Incorporate light elements • Create visual effects • Design for display
Activities	• Light brick integration • Transparent piece usage • Display-worthy builds
Superpower	Showmanship - Making it spectacular

Week 7: Constraint Building

Component	Description
Learning Goals	• Build with limited pieces • Maximize efficiency • Problem-solve creatively

Component	Description
Activities	• Mystery bag challenges • Color-restricted builds • Size-limited challenges
Superpower	Constraint Creativity - Limits inspire innovation

Week 8: Design Competition

Component	Description
Learning Goals	• Compete in building challenges • Present under pressure • Judge others' work
Activities	• Timed competition rounds • Judging workshop • Awards and celebration
Superpower	Competition Readiness - Performing under pressure

Advanced Level

Sophisticated design: Advanced mechanics, artistic building, and professional techniques

Week 1: Pneumatics & Hydraulics

Component	Description
Learning Goals	• Build pneumatic systems • Understand pressure principles • Create powered mechanisms
Activities	• Pneumatic lifts • Hydraulic arms • Pressure-powered machines
Superpower	Fluid Mechanics - Working with pressure

Week 2: Organic Sculpture

Component	Description
Learning Goals	• Create lifelike animals • Master curved surfaces • Build at larger scale
Activities	• Life-size animal sculptures • Human figure building • Organic form mastery
Superpower	Sculptural Mastery - Creating life from bricks

Week 3: Architectural Detailing

Component	Description
Learning Goals	• Add intricate details • Create realistic textures • Master fine-scale work
Activities	• Detailed facade building • Interior decoration • Texture technique exploration
Superpower	Detail Orientation - Perfecting small elements

Week 4: Modular City Planning

Component	Description
Learning Goals	• Design compatible modules • Create city infrastructure • Plan urban environments
Activities	• Street grid system • Modular building sections • City expansion planning
Superpower	Urban Planning - Designing cities

Week 5: Advanced Kinetics

Component	Description
Learning Goals	• Build complex moving sculptures • Create continuous motion • Synchronize multiple movements
Activities	• Multi-axis movement • Synchronized mechanisms • Kinetic art projects
Superpower	Motion Choreography - Orchestrating movement

Week 6: Historical Landmarks

Component	Description
Learning Goals	• Recreate famous buildings • Research architectural history • Build to accurate scale
Activities	• Eiffel Tower or Statue of Liberty • Taj Mahal or Colosseum • Local landmark recreation
Superpower	Historical Recreation - Building history

Week 7: Stop Motion Animation

Component	Description
Learning Goals	• Build poseable figures • Plan animation sequences • Create short films

Component	Description
Activities	• Articulated figure building • Scene construction • Animation production
Superpower	Animation - Bringing builds to life

Week 8: Master Builder Showcase

Component	Description
Learning Goals	• Complete signature build • Document building process • Inspire others
Activities	• Signature MOC completion • Process documentation • Master builder exhibition
Superpower	Signature Style - Developing unique voice